

Association Transformations in Multi-tier Structures

1. Introduction

This paper presents a method for learning transformations over multi-tier representations, with a particular focus on the modification of cross-tier relations (i.e., associations between tiers), which is widely used in linguistics. We consider a specific but linguistically important type of transformations where the nodes remain unchanged from input to output, while only the association relation is modified subject to non-crossing constraints. We formalize these representations as temporally-ordered tiers with a binary association relation, and the learning problem is therefore framed as inferring the which association relations are preserved, deleted, or newly introduced from observed input-output pairs.

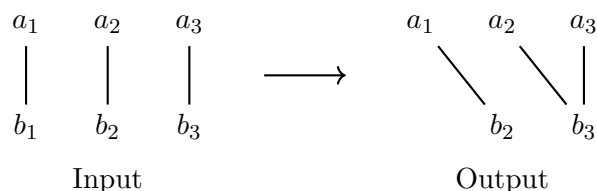


Figure 1: Learning association transformations between tiers.

Such multi-tiered structures have been widely used in linguistic theory, such as tone-syllable associations in autosegmental phonology (Goldsmith, 1976) prosodic or syllable structure (Hayes, 1980; Sagey, 1986). This insight implies that Linguistic information of different aspects are organized across multiple temporally ordered tiers. Associations between tiers encode which elements are linked at a given point in time, thereby capturing relationships across different dimensions of the structure. We refer this type of graph as *autosegmental graphs* (ASGs), which can be viewed as bipartite graphs, where nodes with shared properties are organized into the same tiers. Since each tier is linearly ordered, associations are subject to non-crossing constraint (NCC, Bird and Ellison, 1994; Goldsmith, 1976). The core idea behind autosegmental graphs is that input-output mappings are in fact more naturally characterized by rearranging association lines across tiers rather than changing elements themselves. A large body of theoretical work on African tonal processes has demonstrated the advantages of autosegmental graphs (Hyman, 2009). Computationally, using multi-tiered structures has been shown to reduce the complexity of tonotactic patterns (Jardine, 2016).

Previous work has modeled these structures using finite-state approaches (Wiebe, 1992; Bird and Ellison, 1994; Yli-Jyrä, 2013, 2015). These approaches typically convert structured representations into strings or synchronized tapes to leverage finite-state machinery, or rely on hard-coded mechanisms to represent different types of associations.

In contrast, we propose an approach that learns association transformations between tiers directly over autosegmental graphs using model theory and formal logic. As a preliminary study, we begin with an idealized assumption: tonal processes involve no changes to segmental content, but only the rearrangement of association lines between the underlying representation (UR) and the surface representation (SR). In addition, we assume that only a single process constitutes the learning target.

We show that, by adopting autosegmental graphs within a model-theoretic framework of formal transductions, it is possible to construct a partial order over structures and apply a model-theoretic learner such as the Bottom-Up Factor Inference Algorithm (Chandley et al., 2019; Rawski, 2021). The target output of the learner is a formula that preserves unchanged associations, deletes associations that must be delinked, and introduces newly added associations. Learning proceeds through iterative hypothesis updates while traversing a partially ordered space of logically possible structures.

Furthermore, transformations that appear non-deterministic when modeled over strings can instead be expressed deterministically within this relational framework. To our knowledge, the problem of learning local transformations of association relations in multi-tier structures has not previously been addressed in this form.

2. Preliminaries

2.1. Autosegmental graphs

Definition 1 (Autosegmental Graph) *An autosegmental graph (ASG) is a labeled bipartite graph with multiple linearly-ordered tiers. Equivalently, it can be defined model-theoretically as a relational structure $\mathcal{A} = \langle D, \mathcal{T}, \alpha, \triangleleft \rangle$ where D is a finite domain of elements; each tier $T_i \subseteq D$ is a set of nodes. $\mathcal{T} = \{T_1, \dots, T_n\}$ is a collection of tiers such that $T_1 \cup T_2 \cup \dots \cup T_n = D$ and $T_i \cap T_j = \emptyset$ for all $i \neq j$. $\triangleleft$ is a binary successor relation defined between nodes on the same tier; $\alpha \subseteq D \times D$ is a set of non-crossing association relations across tiers.*

ASGs represent the idea that different aspects of linguistic structure are organized on distinct tiers while progressing simultaneously over time. Each tier encodes a particular type of information, such as tone (pitch information), segmental (tongue shape), syllabic (metrical) structure. Association relations link elements across tiers that are realized at the same temporal position.

Note that there may be multiple tiers in ASGs, provided that the elements on different tiers are disjoint. In most phonological applications, ASGs typically involve two or three tiers. In this work, we assume a two-tiered structure, though the definition remains the same in the general case. In addition, ASGs are subject to the No Crossing Constraint (NCC), which enforces temporal alignment between associations.

Definition 2 (No Crossing Constraint (NCC)) *An ASG needs to obey No Crossing Constraint. Formally, For all $(x, y), (x', y') \in \alpha$, $x \triangleleft x' \Rightarrow (y \triangleleft y') \vee (y = y')$. and vice versa.*

In addition to representing model-theoretic relational structures with graphs, one can use *adjacency matrices* which use binary values to denote whether there is an edge between two nodes. Two nodes are considered *adjacent* if there is an edge connecting them.

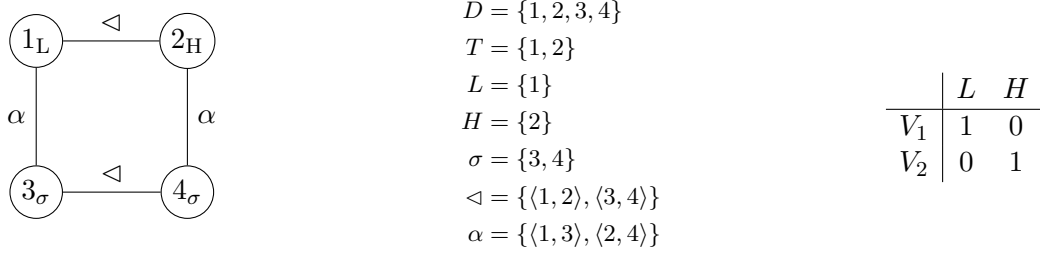
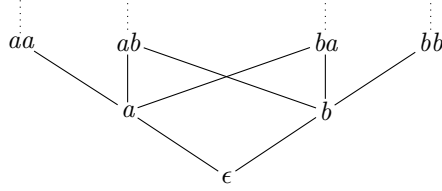


Figure 2: Three representations of a two-tier autosegmental graph.


 Figure 3: Strings over $\Sigma = \{a, b\}$ ordered by the containment relation (non-exhaustive)

Definition 3 (Adjacency matrix) Given an ASG $\mathcal{A} = \langle D, \mathcal{T}, \alpha, \triangleleft \rangle$ and two tiers $T_i, T_j \in \mathcal{T}$, the association relation α can be represented as a binary matrix where $M_\alpha[k, \ell] = 1$ if $(x_k, y_\ell) \in \alpha$ where $x_k \in T_i, y_\ell \in T_j$; otherwise 0.

The motivation for adopting matrices is to establish a framework corresponding to string-based learning. Concepts such as substrings, subfactors, and superfactors are well developed and widely used in grammar inference [NEED REF]. To incorporate ASGs into this framework, we follow the notions of *restriction* and *containment* proposed in Chandlee et al. (2019), adapting them to graph-based representations through the notions of submatrices and matrix containment.

Definition 4 (Submatrix; comparable restriction in Chandlee et al. (2019)) Let matrix A be an $m \times n$ matrix. A matrix B is a submatrix of A if it is formed by taking k ($1 \leq k \leq m$) contiguous rows and v ($1 \leq v \leq n$) contiguous columns from A .

Definition 5 (Containment) An $m \times n$ matrix $A = [a_{ij}]$ is contained in a matrix B (written $A \subseteq B$) if there exists an $m \times n$ submatrix of B , denoted $B' = [b'_{ij}]$, such that for all $i \in [0, m]$ and $j \in [0, n]$, $b'_{ij} \geq a_{ij}$.

The containment relation provides a partial ordering over the space of all model-theoretic structures. As illustrated by the inverted pyramid, more general factors appear lower in the ordering, while more specific factors appear higher positions. This structure provides the foundation for the learning algorithm, which determines whether a structure is contained in another. Importantly, all factors that contain a forbidden structure, i.e., its superfactors, are also forbidden, while all factors contained in a well-formed structure, i.e., its subfactors, are also well formed.

Definition 6 (Neighbor) Given an ASG $\mathcal{A} = \langle D, \mathcal{T}, V, \alpha, \triangleleft \rangle$, a node n is a neighbor of a node m iff $m \triangleleft n \vee n \triangleleft m$.

2.2. Logical Transduction

Using model theory (Strother-Garcia, 2019; Yolyan, 2025; Jardine, 2017), we model a phonological transformation from an underlying representation (UR) to a surface representation (SR) as a relation between input and output structures $\mathcal{A}_i \rightarrow \mathcal{A}_o$. There have been work adopting the same fashion on string-to-string transduction [NEED REF], but the essentially the logic is not restricted to strings and can operate over any model-theoretic representations. In this framework, we use a set of logic formula to derive the output signature with reference to the input signature. Recall from the previous section that we defined an ASG signature as $\mathcal{A} = \langle D, \mathcal{T}, \alpha, \triangleleft \rangle$. Given an input model with this signature, the output signature can be defined in terms of the input structure.

Output	Input	Interpretation
ϕ_{Domain}	$:= \text{TRUE}$	Preserve all input elements
COPY	$:= 1$	Define Output domain the same size as Input
$\phi_{\text{license}}(x)$	$:= \exists y \alpha(x, y)$	Preserve only associated elements
$T'(x)$	$:= T(x)$	Preserve element labels
$\triangleleft'(x, y)$	$\approx \triangleleft(x, y)$	Preserve linear order within-tier
$\alpha'(x, y)$	$:= (\alpha(x, y) \wedge \neg \text{deassoc}(x, y)) \vee \text{reassoc}(x, y)$	Update association relations

Table 1

As shown in Table 1, the transduction defines an output model with the same domain as the input structure, while filtering out unassociated elements. All unary predicates $T(x)$ in the input signature are preserved in the corresponding output predicates $T'(x)$, meaning that an element x has property T in the output iff it has property T in the input. Likewise, the within-tier order relation is preserved. The only change is the association relation α , which is defined such that an association line holds in the output iff either (i) it is present in the input and not removed by **deassoc**, or (ii) it is introduced by **reassoc**.

$$\begin{aligned} \alpha'(x, y) &:= (\alpha(x, y) \quad \wedge \quad \neg \text{deassoc}(x, y)) \quad \vee \quad \text{reassoc}(x, y) \\ \text{output} &= \text{unchanged} \quad - \quad \text{delinked} \quad + \quad \text{newly added} \end{aligned}$$

In this way, the problem of learning association-line transformations can be reduced to learning the logical formulas for **deassoc** and **reassoc**. Since these two predicates are combined through disjunction, two parallel learning processes take place simultaneously, and their outputs are combined to define the final association relation α' .

We demonstrate this process using a common phenomenon in tone languages known as *contour formation*. When two vowels become adjacent, the first vowel (V_1) may be deleted. Since the tone originally associated with V_1 cannot remain unattached, the deletion triggers delinking of T_1 from V_1 and reassociation of T_1 to the following vowel V_2 . As a result, the surviving vowel bears both tones in their original order, which together surface as a contour tone.

From a logical perspective, this transformation can be characterized through the interaction of the **deassoc**, **reassoc**, and ϕ_{license} functions. The **deassoc** operation removes the

association between the deleted vowel and its tone, indicated by the crossed-out line in Figure 4. The **reassoc** operation then links the tone of the deleted vowel (T_1) to the surviving vowel (V_2), indicated by the dotted line in Figure 4. Consequently, T_1 and V_1 are no longer associated and therefore are not licensed by ϕ_{license} (shown in gray in Figure 4). Thus, T_1, T_2 and V_2 remain in the surface representation. If we define the relevant context as $\text{hiatus}(x, y) := V(x) \wedge V(y) \wedge \triangleleft(x, y)$, where x is the first vowel (V_1) and y is the second vowel (V_2), then the transformation can be expressed as follows:

$$\begin{aligned} \text{deassoc} &= \alpha(t, v) \wedge \text{delete_V}(v) \\ \text{reassoc} &= \text{survive_V} \wedge \exists v (\alpha(t, v) \wedge \text{delete_V}(v)) \end{aligned}$$



Figure 4

The **deassoc** and **reassoc** operations are equivalently reflected in the adjacency matrices. Figure [NEED REF] shows the input and output matrices corresponding to the structures in Figure 4. As can be seen, the value corresponding to the association between T_1 and V_1 changes from 1 in the input to 0 in the output, reflecting delinking. Conversely, the value corresponding to T_1 and V_2 changes from 0 to 1, reflecting reassociation. In this way, the autosegmental graph, the adjacency matrix, and the logical expressions provide equivalent description of the same transformation. Formally, **deassoc** and **reassoc** can also be identified by $\text{deassoc} = \{(i, j) \mid \text{UR}_{ij} = 1 \wedge \text{SR}_{ij} = 0\}$ and $\text{reassoc} = \{(i, j) \mid \text{UR}_{ij} = 0 \wedge \text{SR}_{ij} = 1\}$.

2.3. Association Learning

This section follows the framework of using logical transductions to learn phonological mappings in Yolyan (2025), which leverages the entailment relation in a partially ordered space and updates hypotheses about contextual environments. One important aspect of the framework is the use of two sets of k -factors: $V(X)$ and $\hat{V}(X)$, which contain the relevant k -factors associated with the target transformation. $V(X)$ contains the *positive evidence*, namely all k -factors in which the target transformation takes place from the input to the output. In contrast, $\hat{V}(X)$ contains all *negative evidence* where the target transformation does not take place, that is, where the relevant property remains unchanged between the input and output structures.

Yolyan (2025) demonstrates how to learn a string-based input-output mapping that can be written as $A \rightarrow B / C_D$, where the goal of learning is to infer the contextual environment under which the transformation applies. She defines

Definition 7 For every $X \in \{\Phi_F, \varphi_F\}_{F \in \mathcal{F}}$, the corresponding hypothesis space is defined as

$$\mathcal{H}(X) := \bigcup_{x \in V(X)} \downarrow x - \bigcup_{x \in \hat{V}(X)} \downarrow x.$$

In autosegmental phonology, such processes are not always easy to express linearly, since the relevant structure involves relations rather than simple strings. However, the learning question remains the same: what environment triggers the change from 1 to 0, and what environment triggers the change from 0 to 1?

With only a single UR–SR pair, there are infinitely many possible hypotheses. For example, one might hypothesize that all H tones delete their associations, or that all vowels do. However, once more data points are observed, counterexamples eliminate incorrect generalizations.

The learning task, therefore, is to determine the surrounding environment of a particular cell value—either 1 or 0—that triggers the change.

$$\begin{array}{ccc} ? & ? & ? \\ ? & \mathbf{0} & ? \\ ? & ? & ? \end{array} \rightarrow \begin{array}{ccc} ? & ? & ? \\ ? & \mathbf{1} & ? \\ ? & ? & ? \end{array}$$

Input: $D = \{(I_1, O_1), \dots, (I_n, O_n)\}$
Output: A hypothesis H
 $H, V^+, V^-, P \leftarrow \emptyset$
 $Q \leftarrow \{\alpha(x, y)\}$
foreach $(I, O) \in D$ **do**
 $V^+ \leftarrow V^+ \cup (\alpha_I \setminus \alpha_O)$
 $V^- \leftarrow V^- \cup (\alpha_I \cap \alpha_O)$
while $Q \neq \emptyset$ **do**
 $c \leftarrow Q.\text{Dequeue}()$
 if $\forall g \in V^+, c \sqsubseteq g$ **then**
 if $\forall g \in V^-, c \not\sqsubseteq g$ **then**
 $H \leftarrow H \cup \{c\}$
 else
 foreach $s \in \text{NEXTSUPFACT}(c)$ **do**
 if $\forall p \in P, s \not\sqsubseteq p$ **then**
 $Q.\text{Enqueue}(s)$
 else
 $P \leftarrow P \cup \{c\}$
return H

3. Case Study

3.1. Contour Formation

We begin with an idealized assumption: the tonal process involves no change to the segmental string, and the only modifications between the underlying representation (UR) and the surface representation (SR) are changes in association lines, namely DEASSOCIATION and REASSOCIATION. Although many phonological processes also involve segmental deletion, insertion, or feature changes, we restrict the current discussion to association-line modifications in order to isolate the learning problem.

The first step is to identify exactly what structural modification has taken place between the UR and the SR. Since the segmental and tonal tiers remain constant, the comparison is made directly over the association relation matrix.

For each cell in the association matrix:

That is, if an association present in the UR disappears in the SR, the operation is classified as deassociation; if a new association appears in the SR that was absent in the UR, the operation is classified as reassociation.

In Estoko, both operations are attested: one association line is removed while another is added. Therefore, the learner must infer both the deassociation environment and the reassociation environment separately. (maybe not maybe they can start together)

Hypothesis space for deassociation. Given a set of observed deassociation sites, the learner constructs candidate hypotheses describing the structural environments in which an association changes from 1 to 0. Each hypothesis corresponds to a local structural condition under which $\alpha(x, y)$ is removed.

The search proceeds from simple to more complex descriptions. At each step, the learner considers increasingly rich local configurations (factors) until a hypothesis is found that captures all and only the observed deassociation sites.

The most general hypothesis removes all associations:

Hypothesis 1: $1 \rightarrow 0$

This hypothesis is immediately rejected if there exist associations in the data that remain unchanged.

4. Conclusion

This paper has argued that vowel-deletion-induced tonal processes can be uniformly captured as logical transductions over autosegmental representations (ARs). The observed typological variation follows from modifying a single relation, α , through the interaction of deassoc and reassoc.

A limitation of the current analysis is that the transduction relies on quantification to capture across-tier dependencies. An important direction for future work is to determine whether these processes can be simplified in a quantifier-free (QF) process. This work also extends naturally to learning over ARs. Recent work by Yolyan (2025) demonstrates that logical transductions are learnable by using a bottom-up factorial inference algorithms (Chandlee et al., 2019). This opens up more discussions about how linguistically meaningful representations can facilitate grammar inference.

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